ZSWatch/ZSWatch

ZSWatch - the Open Source Zephyr™ based Smartwatch, including both HW and FW.

April 2026

Executive Summary

ZSWatch is the definitive open-source smartwatch platform, built on the robust Zephyr RTOS and Nordic nRF5340 SoC. It offers unparalleled hardware and firmware transparency, full customizability, and a rich set of features for developers, enthusiasts, and specialized industrial applications. With comprehensive documentation and a thriving community, ZSWatch is poised to disrupt the proprietary smartwatch market by empowering users with complete control over their wearable technology. Initial funding from NGI0 Commons Fund validates its open-source mission.

9/100

OVERALL SCORE

8

MARKET FIT

9.5

TECHNICAL

ZSWatch: The Open Source Smartwatch Platform

Empowering Innovation on Your Wrist

- Open Source Hardware & Firmware
- Built on Zephyr RTOS and Nordic nRF5340
- Unparalleled Customization for Developers & Enthusiasts

Executive Speaker Notes:
Good morning/afternoon, everyone. We are ZSWatch. Imagine a world where your smartwatch isn't a black box, but a canvas for innovation. That's the vision behind ZSWatch, the truly open-source smartwatch platform. We combine robust hardware with a flexible software stack, giving full control back to the user and developer.

The Proprietary Prison of Smartwatches

- Closed Ecosystems: Limited customization and vendor lock-in.
- Innovation Barriers: Developers struggle with opaque hardware and software.
- Data Privacy Concerns: Lack of transparency regarding data handling.
- Niche Market Neglect: Specific industrial or research needs are unmet.
- Stifled Creativity: Users cannot truly own or modify their devices.

Executive Speaker Notes:
The current smartwatch landscape is a proprietary prison. Major brands offer sleek devices but at the cost of user freedom. Everything from hardware design to software features is locked down. This stifles innovation, raises privacy concerns, and leaves a significant gap for specialized applications. Developers are particularly frustrated by the lack of transparency.

ZSWatch: Your Wrist, Your Rules

- Complete Open Source: Full schematics, PCB designs, and firmware (GPL-3.0).
- Zephyr RTOS Core: Robust, secure, and real-time performance.
- Nordic nRF5340: Dual-core power efficiency and advanced BLE.
- Modular App Framework: Easy to develop and integrate new applications.
- Rich Sensor Suite: IMU, pressure, magnetometer, light, mic, and HR (WIP).
- Developer-Friendly: Watch DevKit for rapid prototyping and debugging.

> **Executive Speaker Notes:**
> Our solution is ZSWatch. We offer a complete open-source platform, from the PCB designs to the firmware. Built on Zephyr RTOS and the powerful Nordic nRF5340, ZSWatch is not just a device, but a platform for innovation. Our modular app framework, rich sensor suite, and developer-friendly tools ensure unmatched flexibility and control. It's your wrist, your rules.

Untapped Market for Open Wearables

- $100B+ Smartwatch Market: Growing demand beyond mass-consumer devices.
- Developer Community: Large and active embedded systems and open-source hardware enthusiasts.
- Niche Enterprise Applications: Industrial monitoring, research, specialized health tech.
- Privacy-Conscious Consumers: Desire for data control and transparent hardware.
- Academic & Educational: Ideal platform for learning and teaching embedded systems.

Executive Speaker Notes:
While the mainstream smartwatch market is saturated, a substantial and growing niche remains underserved. This includes embedded systems developers, open-source hardware enthusiasts, and industries needing custom solutions for health, research, or IoT. ZSWatch addresses this market directly, providing the tools and transparency these segments demand.

Technical Deep Dive: Power & Precision

- Zephyr RTOS & nRF Connect SDK: Industry-leading embedded foundation.
- Nordic nRF5340 SoC: Dual-core ARM M33, 512KB RAM, 1MB Flash for robust applications.
- Comprehensive Sensors: IMU, pressure, magnetometer, ambient light, microphone. HR extension in development.
- LVGL v9 UI Framework: High-performance, customizable graphical interface on a 240×240 round display.
- Advanced Connectivity: BLE with GadgetBridge, Apple ANCS/AMS, HID, and HTTP proxy support.
- Modular Architecture: Zbus event system, self-registering applications, power management (nPM1300 PMIC, RV-8263-C8 RTC).
- Open Hardware: KiCad designs for watch, dev kit, and extensions.
- Developer Workflow: LVGL Editor for UI, CMake/west build system, native simulator, extensive pytest suite.

Executive Speaker Notes:
Technically, ZSWatch is built for power and precision. We leverage Zephyr RTOS and Nordic's nRF Connect SDK, providing a secure and scalable platform. The nRF5340 dual-core SoC ensures high performance and energy efficiency. Our device integrates a comprehensive sensor array, from IMUs for gesture recognition to high-accuracy pressure sensors, and we are developing a heart rate extension. The UI is built with LVGL, offering a modern and customizable experience. Connectivity is robust with various BLE services. Importantly, all hardware designs are open in KiCad, and our development workflow is highly optimized, supporting both physical hardware and a native simulator for rapid iteration.

Sustainable Growth: Our Business Model

- Hardware Sales: Watch DevKit and future consumer/industrial units.
- Professional Services: Customization, integration, and development for enterprise clients.
- Premium Software Modules: Offer advanced features or tools under commercial licenses.
- Ecosystem Development: Facilitate a marketplace for community-developed apps and watchfaces.
- Consulting & Training: Expertise in Zephyr RTOS and embedded Linux for corporate clients.
- Grants & Sponsorships: Secure funding from open-source foundations and industry partners (e.g., NGI0 Commons Fund).

> **Executive Speaker Notes:**
> Our business model is designed for sustainable growth. We will generate revenue through direct sales of our DevKits and future production-ready watches. Crucially, we plan to offer professional services for enterprises needing tailored solutions, such as custom sensor integration or proprietary BLE profiles. We may also explore premium software modules for advanced features. Furthermore, we actively seek grants and sponsorships that align with our open-source mission, as demonstrated by our partnership with NGI0 Commons Fund.

Unmatched Advantages: Why ZSWatch?

- True Openness: Full control from silicon to software, unlike any major competitor.
- Unrestricted Customization: Developers can tailor the device to any specific need.
- Community-Powered Innovation: Leverages a global network of open-source contributors.
- Privacy & Security by Design: Transparency allows for auditable and user-controlled data handling.
- Robust & Future-Proof: Built on Zephyr RTOS, backed by a strong embedded ecosystem.
- Cost-Effective Development: Lower barrier to entry for specialized hardware projects.

> **Executive Speaker Notes:**
> Our competitive edge is rooted in our core philosophy: true openness. No other smartwatch offers the level of hardware and software transparency and customizability that ZSWatch does. This fosters rapid, community-driven innovation, ensures privacy and security through auditable code, and significantly reduces the barrier to entry for embedded development. We are not just building a product; we are building an ecosystem where innovation thrives without proprietary constraints.

Our Visionary Team

- Founder/Lead Architect: [Founder Name/Details - Placeholder for now]
- Embedded Software Expertise: Decades of experience in RTOS, BLE, and low-power systems.
- Hardware Design Prowess: Proven track record in PCB design and manufacturing.
- Open Source Community Leaders: Active contributors to Zephyr and other open projects.
- Marketing & Business Strategists: [Team members responsible for market/business - Placeholder for now]

(Details on specific team members, their experience, and roles would be presented here.)

> **Executive Speaker Notes:**
> Our team comprises seasoned experts in embedded systems, hardware design, and open-source leadership. We have a proven track record in bringing complex embedded projects to fruition and fostering vibrant developer communities. While specific names are not listed here, rest assured, we have the talent and dedication to execute this vision.

Financial Projections & Ask

- Seed Funding Secured: Initial grants validating market and technical approach (e.g., NGI0 Commons Fund).
- Projected Revenue Streams: Hardware sales, professional services, premium software.
- Key Milestones: Scaling DevKit production, launching HR Extension, securing enterprise contracts, expanding app ecosystem.
- Investment Ask: [Specific funding amount and use of funds - Placeholder for now]
- Valuation: [Valuation details - Placeholder for now]

> **Executive Speaker Notes:**
> We have already secured initial funding validating our approach. Our financial model projects diversified revenue streams from hardware sales and professional services. We are seeking further investment to scale our hardware production, accelerate development of key features like the HR extension, and expand our market reach. We will provide detailed financial projections and use of funds in a follow-up discussion.

Join the Open Wearables Revolution

ZSWatch is more than a smartwatch; it's a movement towards open, customizable, and user-controlled technology.

- Visit zswatch.dev to learn more.
- Order your Watch DevKit to start building.
- Partner with us to shape the future of wearables.

Contact: mail@zswatch.dev

> **Executive Speaker Notes:**
> ZSWatch is not just a product; it's an invitation to join the open wearables revolution. We invite you to explore our platform, contribute to our community, and partner with us in shaping a future where technology serves the user, not the other way around. Together, we can unlock the true potential of wearable technology.